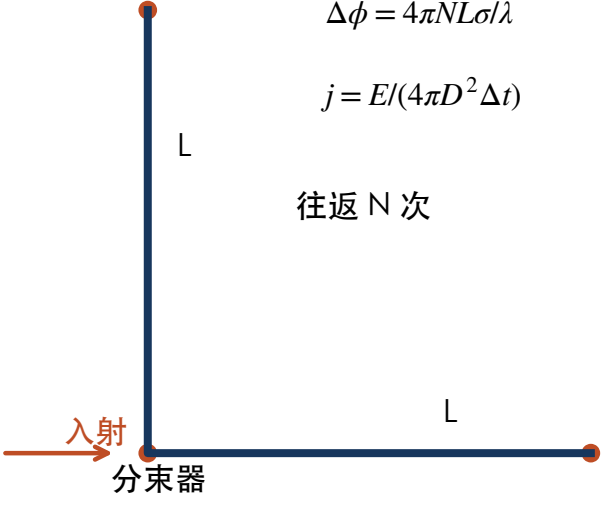
$$\Delta\phi = 4\pi NL\sigma/\lambda$$

$$j = E/(4\pi D^2 \Delta t)$$